EDA Case Study Presentation

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Project overview

We looked through different sets up data on loan applications to understand why applicants were defaulting, specifically delving into behavior and habits in intent.

Understanding the Data

Importing Data sets

-Application: contains all the information of the client at the time of application. The data is about whether a client has payment difficulties.

-Previous: contains information about the client's previous loan data. It contains the data whether the previous application had been Approved, Cancelled, Refused or Unused offer.

```
from google.colab import drive
drive.mount('/content/drive/')
```

```
Mounted at /content/drive/
```

```
!ls "/content/drive/MyDrive/Colab Notebooks"
```

```
import pandas as pd

app = pd.read_csv("/content/drive/MyDrive/Colab Notebooks/bootcamp_eda_case/application_data.csv.zip", compression = "zip")
prev = pd.read_csv("/content/drive/MyDrive/Colab Notebooks/bootcamp_eda_case/previous_application.csv.zip", compression = "zip")
```

Education

Looking at education as a possible indicator of defaulting (TARGET)

```
education = app.groupby("NAME_EDUCATION_TYPE")["TARGET"].mean()
```

```
education * 100
```

NAME_EDUCATION_TYPE	TARGET
Academic degree	1.829268
Higher education	5.355115
Incomplete higher	8.484966
Lower secondary	10.927673
Secondary / secondary special	8.939929

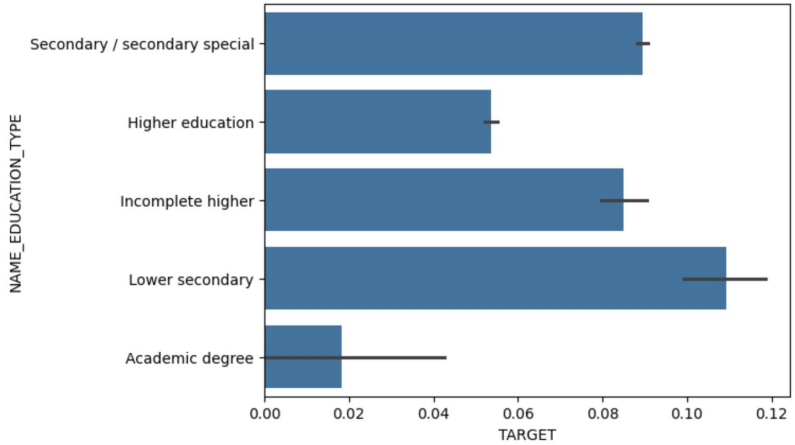
dtype: float64

```
import matplotlib.pyplot as plt
import seaborn as sns

app.groupby("NAME_EDUCATION_TYPE")["TARGET"].mean()

sns.barplot(data=app, x="TARGET", y="NAME_EDUCATION_TYPE")
```

```
<Axes: xlabel='TARGET', ylabel='NAME_EDUCATION_TYPE'>
```

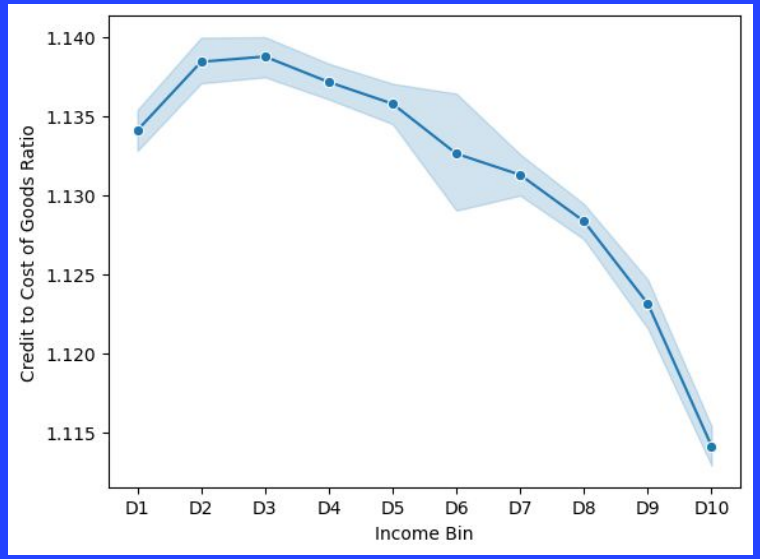


Conclusions from Education

These charts show a significant increase in default likelihood with less education obtained.

In the barplot it shows the confidence intervals with the “--” on the ends, which indicates that very little of the data included applicants with an academic degree, whereas there was a lot of data of those with higher education of some sort.

Financing Structure by Income



Data Analysis

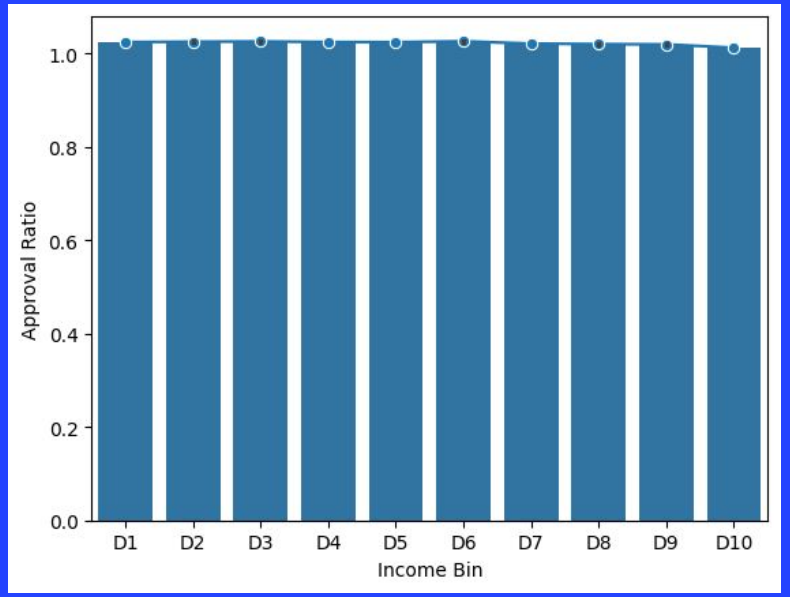
- Current Application Dataset
- Demand Ratio:
  - Credit Amount : Cost of Goods
- Grouped applicants into Income Buckets (Min: $25,650; Max: 117M)
-

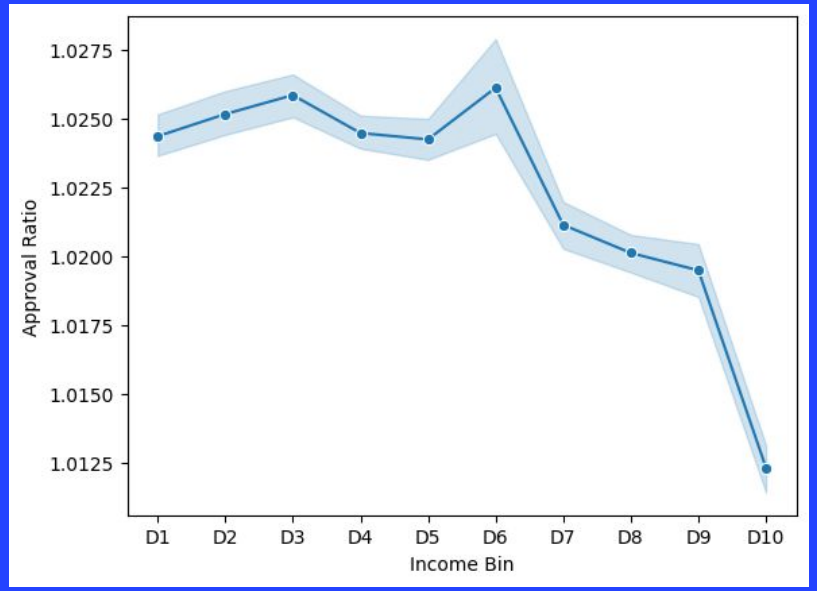
Cleaning

- Controlled for null values & extreme ratios → removed Revolving Credit & removed outliers

Observation → Demand Ratio tends to go down as income bin increases

Approval Ratio by Income Bucket



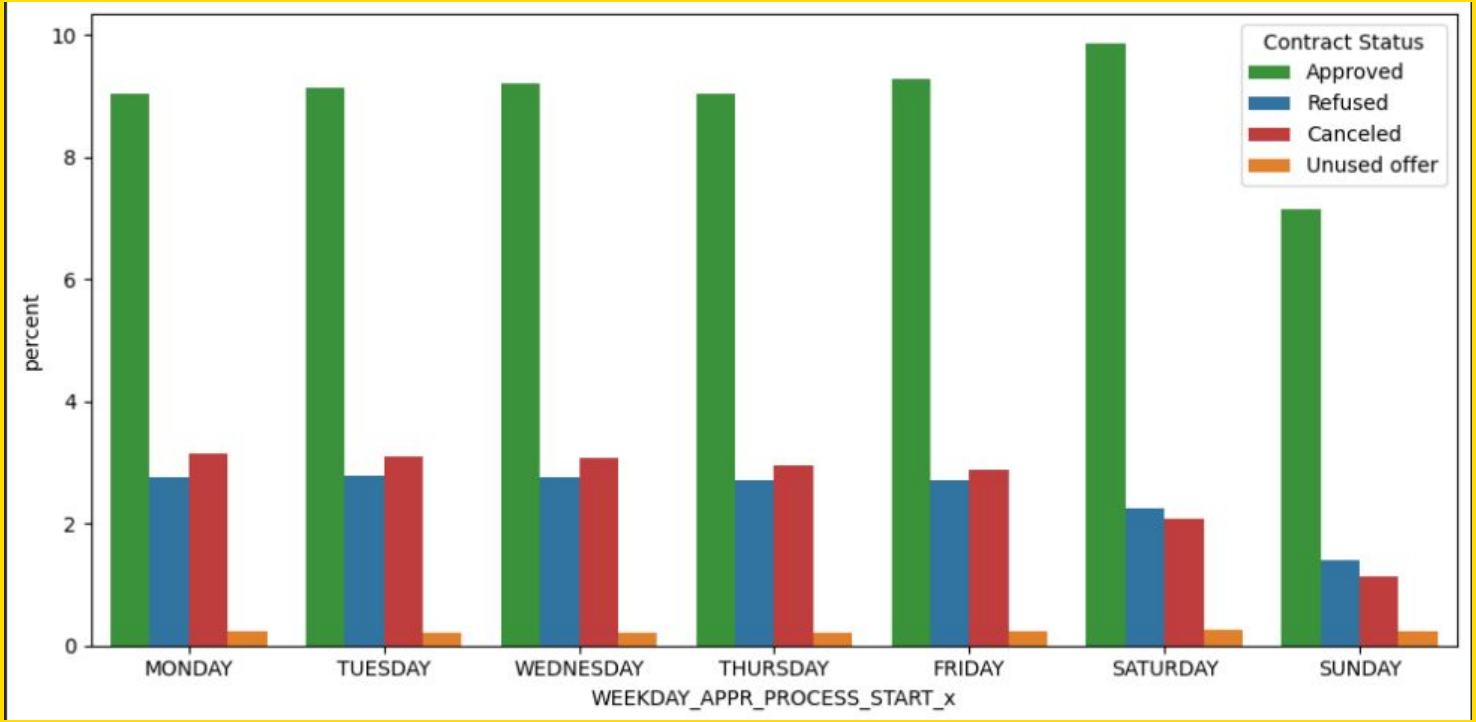


- Previous Applicant Dataset
- Same information
- Both charts imply different results
- Observation→ Subtle differences in approval ratio by income bucket

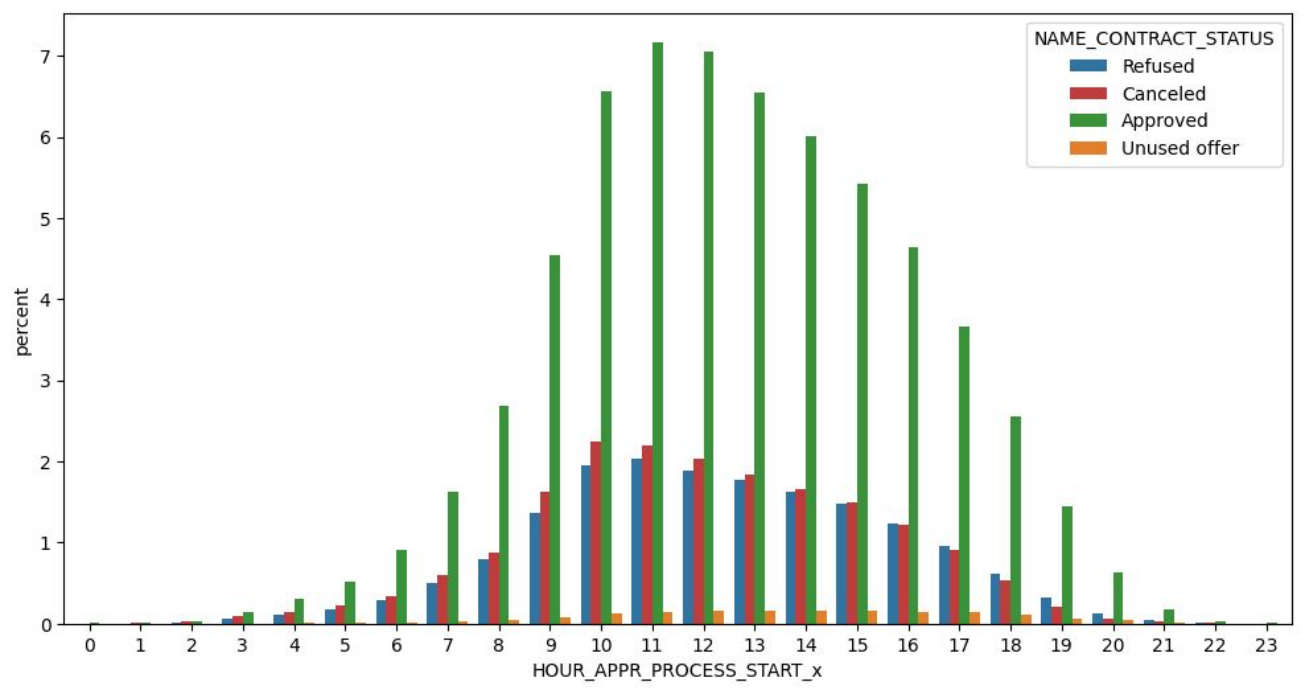
Conclusions from Financing

- Minimal variation in approval ratios across income tiers → income is not a primary determinant of financing structure.
- Slightly lower ratios at higher incomes suggest borrower request behavior, not lender constraint, drives differences.
- No strong evidence of systematic income-based scaling in approvals.
- Conclusion: Financing structure appears largely behavior-driven rather than structurally constrained by income.

Application Start Times - Weekdays



Application Start Times - Hours



Conclusions from Application

- Most applications that start on the weekends get approved, more specifically on Saturdays.
- Approvals based on weekday start times are consistent, with few variations.
- If an applicant starts their form in the late morning to mid afternoon (from 10am to 4pm), they are more likely to get approved.
- More applicants cancel their loan applications than wait for it to be refused.